Research Proposal

Rational Design of Ion-Migration Suppressing Dopants in Halide Perovskites

A First-Principles and Machine-Learned Interatomic Potential Screening Framework for Intrinsic Stability

Candidate: Eric
Programme: Natural Sciences (Materials Science & Chemistry)
Institution: University of Cambridge

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One-sentence summary. This project converts perovskite stability design from empirical trial-and-error into a white-box computational screening problem: it ranks ~50 candidate "pinning" dopants by their first-principles effect on the iodide-vacancy migration barrier, attributes every effect to a physical mechanism, and introduces the *pinning radius* — the spatial range of a dopant's protection — as a computable design metric, the whole campaign accelerated four orders of magnitude by an actively-learned machine-learned interatomic potential.

Abstract

Halide perovskite solar cells have surpassed 27% certified single-junction efficiency, yet operational instability remains the decisive barrier to commercialisation. The instability is rooted in an intrinsic contradiction: the soft, ionic, low-formation-energy lattice that grants perovskites their exceptional optoelectronic properties simultaneously permits facile ion migration (activation energies $E_a \approx$ 0.1–0.6 eV), which acts as the mechanistic hub linking hysteresis, halide segregation, interfacial corrosion, and field-driven degradation. Raising E_a — via lattice strain management or A-site "pinning" cations — demonstrably extends device lifetime, but dopant discovery remains empirical: the two computational studies published to date cover fewer than five dopants at a single lattice site, and the field's authoritative reviews concede that no strategy can yet specify *which* defect species it immobilises. This proposal develops a five-layer computational framework — Born–Oppenheimer potential energy surfaces, harmonic transition-state theory, climbing-image nudged elastic band calculations, density functional theory with charged-defect corrections, and *per-charge-state* fine-tuned MACE machine-learned interatomic potentials (MLIPs) — to compute the dopant-induced shift ΔE_a for iodide-vacancy migration across a combinatorial space of ∼50 substitutional and interstitial dopants. The deliverables are (i) the first *mechanism-annotated* dopant ranking at this scale, (ii) the *pinning radius*: a new metric extracted in MLIP-scale supercells that converts each dopant's spatial range of action into the minimum concentration required for lattice-wide protection, and (iii) an open-source, reusable screening pipeline. Every ranked prediction is gated behind four experimental and computational anchors, and the project is feasible within a twelve-month timeline on standard university HPC resources.

Contents

1 Background and Motivation

1.1 The intrinsic-instability paradox of halide perovskites

Metal halide perovskites (ABX_3; A = MA^+, FA^+, Cs^+; B = Pb^{2+}; X = I^-, Br^-) combine strong absorption, long carrier lifetimes, and high defect tolerance with room-temperature solution processability. These virtues, however, derive from precisely the material properties that undermine stability: a soft, strongly ionic lattice and a low formation energy. The Goldschmidt tolerance factor of the workhorse compositions ($FAPbI_3$, $MAPbI_3$) sits at the edge of the stable window, so modest perturbations of temperature, humidity, or bias drive transformation to photoinactive phases. The favourable and unfavourable properties are two faces of the same physics; the instability therefore cannot be eliminated, only suppressed to an engineering-acceptable level.

1.2 Ion migration as the mechanistic hub

Among all degradation channels, ion migration occupies a privileged causal position. Density functional theory identifies vacancy-mediated iodide hopping along the edge of the PbI_6^{4-} octahedron as the lowest-energy transport mechanism, with activation energies of only 0.1–0.6 eV (Eames et al., 2015) — low enough for migration at room temperature under the built-in field of an operating device. Migrating ions produce current–voltage hysteresis on short timescales and, on long timescales, interfacial ion accumulation, electrode corrosion (e.g. AgI formation), photoinduced halide segregation in wide-bandgap compositions, and stoichiometric drift (Thiesbrummel et al., 2026). Critically, whereas moisture, oxygen, and UV exposure can be excluded by encapsulation, the operating electric field cannot: a device degrades by ion migration merely by working. Suppressing ion migration is therefore not one stability strategy among many; it is the strategy on which all others depend.

1.3 The exponential leverage of the activation energy

Transition-state theory places E_a in the exponent of the hop rate,

$$\Gamma = \nu_0 \exp\left(-\frac{E_a}{k_B T}\right), \tag{1}$$

so that at 300 K an increase of $\Delta E_a = 0.20$ eV suppresses the migration rate by a factor of $\exp(0.20/0.0257) \approx 2.4 \times 10^3$. No other single materials parameter offers comparable leverage over operational stability. Recent experiments confirm the principle: relaxation of residual tensile strain at the buried hole-transport-layer interface of inverted (p–i–n) cells raised the measured E_a from 0.47 eV to 0.63 eV and conferred resilience to reverse-bias stress (Du et al., 2026), while partial substitution of guanidinium (GA^+) — which forms seven effective hydrogen bonds to the PbI_6^{4-} framework versus three for MA^+ (Lee et al., 2022) — yielded devices retaining 90% of initial efficiency after 1200 h of continuous illumination (Li et al., 2022).

2 State of the Art and the Research Gap

2.1 What is established

Four bodies of evidence now frame the problem:

1. **Strain management.** Residual tensile strain — arising from thermal-expansion mismatch with the substrate and from photoinduced lattice expansion — elongates and weakens Pb–X

bonds, lowering defect formation energies and E_a; conversion to mild compressive strain raises the barrier to both migration and halide segregation (Muscarella et al., 2020; Liu et al., 2021; Xue et al., 2020).

2. **A-site pinning cations.** Large cations such as GA^+ anchor the halide sublattice through extended hydrogen-bond networks, raising E_a as measured by temperature-dependent ionic conductivity (Li et al., 2022; Lee et al., 2022).

3. **Multivalent and multi-site doping.** Trace interstitial multivalent cations electrostatically immobilise halides (Zhao et al., 2022), and computational surveys have examined dopants across A, B, and X sites.

4. **Computational precedents.** Two 2025 studies establish the computational baseline for exactly this system. Tyagi et al. (2025) trained charge-state-specific machine-learned force fields for iodide vacancy and interstitial migration in bulk $CsPbI_3$, showing that the defect charge state alters mobility by an order of magnitude. Arber, Mocanu and Islam (2025) combined DFT and MLIP molecular dynamics ($\sim$80 ns) to show that B-site substitution (Sn^{2+}, Ba^{2+}, Cu^{2+}) does *not* materially suppress iodide transport in γ-$CsPbI_3$. These works validate the MLIP methodology for halide-perovskite defect migration — but their coverage ($\leq$3 dopants, one lattice site, no mechanistic attribution) leaves the screening question open, and the B-site null result sharpens it: the promising chemical space is precisely the A-site pinning and interstitial multivalent chemistry this project targets.

2.2 The gap: a black-box search without mechanistic labels

Despite these advances, dopant discovery remains empirical. The search confronts three compounding obstacles:

The black-box optimisation dilemma. *Input space:* dopant species $\times$ lattice site $\times$ concentration — combinatorially 10^3–10^4 options. *Evaluation cost:* one candidate = weeks of synthesis plus $\sim$1000 h of ageing tests. *Gradient information:* none — current experiments reveal *that* E_a rose, not *why*.

The authoritative review of the field is explicit on this point: the vast majority of mitigation strategies fall short of pinpointing the specific defect species that are immobilised or suppressed (Lee et al., 2022), and the existing computational studies cover fewer than five dopants at a single lattice site with no mechanistic labels (Tyagi et al., 2025; Arber et al., 2025). A linear, one-candidate-at-a-time experimental search over an exponentially large space, with no mechanistic feedback, is the precise definition of the bottleneck this project removes.

3 Aims and Objectives

Overarching aim. To replace empirical dopant discovery with rational, mechanism-resolved computational screening, by establishing ΔE_a — the dopant-induced shift in the iodide-vacancy migration barrier — as a computable, experimentally-anchored proxy for intrinsic stability.

Objective 1 (Method validation). Reproduce four anchors with the computational pipeline: (a) the literature range $E_a \approx 0.1$–0.6 eV for iodide-vacancy migration in the undoped lattice (Eames et al., 2015); (b) the charge-state ordering of defect mobilities — in particular the order-of-magnitude separation between V_I^+ and V_I^0 — reported by Tyagi et al. (2025), which serves as a like-for-like computational anchor; (c) the sign and magnitude of the GA^+-induced ΔE_a; (d) the strain–E_a correlation (tensile lowers, compressive raises).

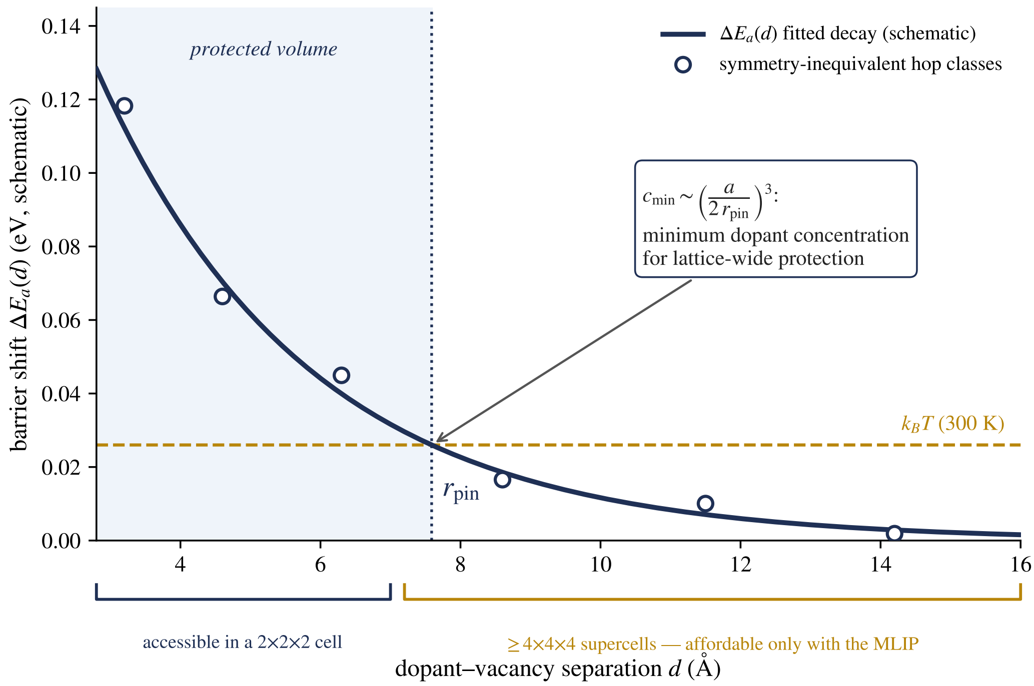

Figure 1: Definition of the pinning radius (schematic). ΔE_a is computed for symmetry-inequivalent hop classes at increasing dopant–vacancy separation d; the fitted decay crosses the thermal scale k_BT at r_{pin}, the radius of the protected volume. The decay tail lies beyond the reach of ranking-scale $2 \times 2 \times 2$ cells and is accessible only with the MLIP; r_{pin} converts directly into the minimum dopant concentration for lattice-wide protection, connecting cell-scale simulations to the trace doping levels used experimentally.

Objective 2 (High-throughput screening). Compute mechanism-annotated ΔE_a values for $\sim$50 candidate dopants spanning A-site substitution (GA^+, dimethylammonium, acetamidinium), B-site substitution (Sr^{2+}, Ca^{2+}, Bi^{3+}, La^{3+}) — retained deliberately as a *negative-control class* anchored to the published null result of Arber et al. (2025) — X-site substitution (Cl^-, SCN^-), and interstitials (K^+, Rb^+, multivalent cations), each over symmetry-inequivalent migration paths binned by dopant–vacancy distance.

Objective 3 (Mechanistic resolution). For every screened dopant, extract the saddle-point atomic configuration and quantify the pinning mechanism — hydrogen-bond count, Pb–I bond-length changes, and Bader charge redistribution — thereby classifying dopants into hydrogen-bond-anchoring, electrostatic-trapping, and channel-blocking types, and defining a *pinning radius* from the decay of ΔE_a with dopant–vacancy separation (Fig. 1). Because the decay tail extends beyond the $\sim$7 Å separations accessible in a $2 \times 2 \times 2$ cell, pinning radii are extracted in $\geq 4 \times 4 \times 4$ supercells ($\geq$700 atoms) — computationally trivial for the fine-tuned MLIP and precisely where MLIP acceleration is irreplaceable. The small-cell campaign supplies the *ranking*; the large-cell campaign supplies the *radius*; and the radius converts cell-scale dopant fractions into the trace concentrations used experimentally (Zhao et al., 2022). Finally, every top-ranked candidate is screened for *electronic benignity*: because the perovskite band edges are built almost entirely from the Pb–I framework, most A-site and interstitial dopants are electronically innocent, but candidates whose density of states shows dopant-induced mid-gap levels — the documented failure mode of Bi^{3+} — are demoted regardless of their ΔE_a rank.

Objective 4 (Open infrastructure). Release the full pipeline (structure enumeration, MLIP-accelerated CI-NEB farm, active-learning loop, analysis) as documented open-source software.

4 Theoretical Framework

The methodology rests on five theory layers, each of which repairs the deficiency of the layer above it.

4.1 Layer 1: Born–Oppenheimer potential energy surface

The mass separation between nuclei and electrons permits integration over electronic degrees of freedom, defining a potential energy surface (PES) $E(\mathbf{R}_1, \ldots, \mathbf{R}_N)$ on which stable configurations are local minima, migration transition states are first-order saddle points (Hessian with exactly one negative eigenvalue), and $E_a = E_{\text{saddle}} - E_{\text{minimum}}$ along the minimum-energy path (MEP). In this language, a pinning dopant is precisely a local reshaping of the PES: deepening the valley or raising the pass.

4.2 Layer 2: Harmonic transition-state theory

Assuming Boltzmann equilibrium in the reactant basin, a dividing surface at the saddle, and no recrossing, the hop rate takes the Vineyard form

$$\Gamma \;=\; \frac{\prod_{i=1}^{3N} \nu_i^{\text{min}}}{\prod_{i=1}^{3N-1} \nu_i^{\text{saddle}}} \exp\!\left(-\frac{E_a}{k_B T}\right), \tag{2}$$

in which the prefactor varies by a factor of 2–3 across chemically similar dopants while the exponential varies by orders of magnitude. This licenses a screening strategy that ranks candidates by E_a alone. The harmonic approximation is justified here since $E_a/k_BT \approx 20$ (deep-well limit), and quantum tunnelling is negligible for an ion as heavy as iodide.

Halide perovskites are, however, strongly anharmonic, with double-well zone-boundary phonons and pronounced dynamic disorder; static 0 K barriers and 300 K effective barriers can therefore differ through entropic and prefactor contributions (Meyer–Neldel compensation). Two safeguards address this. First, the screening observable is the *difference* ΔE_a between doped and undoped lattices, in which anharmonic and entropic corrections cancel to first order — the ranking is more robust than any absolute barrier. Second, for the top-ranked candidates the harmonic-TST picture is cross-checked by transition-state-theory-free molecular dynamics (Section 5.3).

4.3 Layer 3: Saddle-point location by CI-NEB

The climbing-image nudged elastic band method (Henkelman, Uberuaga and Jónsson, 2000) discretises the path between two minima into images coupled by virtual springs, with force projection — spring forces retained only parallel to the path, true forces only perpendicular — preventing corner-cutting and image sliding. The highest-energy image then climbs against the parallel component of the true force to converge exactly onto the saddle point. CI-NEB (rather than plain NEB) is mandatory for quantitative E_a.

4.4 Layer 4: DFT reference energies

Kohn–Sham DFT provides the ground-truth PES at $\mathcal{O}(N^3)$ cost. All production cells use the room-temperature photoactive orthorhombic γ-phase of $CsPbI_3$ (consistent with Arber et al., 2025), whose broken symmetry is handled by explicit enumeration of inequivalent hop directions. Four system-specific corrections are essential: (i) spin–orbit coupling for Pb is included in electronic-structure checks, while migration barriers — governed chiefly by ionic electrostatics and bond strength — are computed at the PBE(+D3) level after explicit SOC-sensitivity tests;

(ii) the dominant migrating defect is the positively charged iodide vacancy V_{I}^{+}, requiring finite-size electrostatic corrections of the Freysoldt–Neugebauer–Van de Walle type (Freysoldt et al., 2009), the neglect of which explains ∼0.1 eV discrepancies between literature values; because FNV is derived for equilibrium point defects, its path-dependence along the MEP is monitored through supercell-size tests; (iii) supercell convergence ($\geq 2 \times 2 \times 2$, ∼96 atoms for the ranking campaign, with extrapolation tests) controls spurious image–image interactions; (iv) PBE self-interaction error can artificially delocalise charge around charged defects, so the top-three candidates receive HSE(+SOC) single-point spot checks — whose density of states doubles as the electronic-benignity screen of Objective 3, one calculation serving two gates — before experimental hand-off. Method development is performed first on the fully inorganic $CsPbI_3$ lattice to decouple methodology from the organic-cation orientational degrees of freedom, which are subsequently sampled by averaging over several cation orientations.

4.5 Layer 5: MLIP acceleration with active learning

A single CI-NEB (∼7 images × ∼100 ionic steps) costs thousands of core-hours at the DFT level; the screening campaign requires ∼2000 such paths, which is infeasible. The MACE equivariant message-passing architecture (Batatia et al., 2022), via foundation models pre-trained on $\sim 10^6$ Materials Project configurations (Batatia et al., 2024), reduces force-evaluation cost by four orders of magnitude. Its built-in $E(3)$-equivariance guarantees exact rotational behaviour of energies and forces by construction, and message passing extends the effective receptive field beyond the local cutoff. The starting checkpoint will be selected among MACE-MP-0, MACE-MPA-0 and OMat24-trained models (Barroso-Luque et al., 2024) by validation against the migration-barrier benchmark of Bheemaguli, Xiao and Sai Gautam (2025), which identifies MACE-family models as the most accurate foundation MLIPs for NEB barriers while confirming that MPtrj-trained models systematically *underestimate* barriers through PES softening (Deng et al., 2025).

The central methodological risk — and its remedy. Foundation MLIPs are trained predominantly on near-equilibrium configurations, whereas saddle points are far-from-equilibrium structures lying precisely in the model's extrapolation regime — and softening biases zero-shot barriers systematically low (Deng et al., 2025). Zero-shot barriers are therefore never ranked: they serve only as *path seeding*, a role in which foundation-model images outperform conventional interpolation in >71% of NEB setups (Bheemaguli et al., 2025). The remedy is a Bayesian active-learning loop: committee-variance uncertainty flags the configurations the model does not know (saddle-point regions are flagged almost by construction); DFT single points are computed only there; the model is fine-tuned; the loop repeats until the validation error satisfies $|E_a^{\mathrm{MLIP}} - E_a^{\mathrm{DFT}}| < 0.03$–$0.05$ eV. Each expensive DFT call is thereby spent where its information gain is maximal.

Charge-state representation in the MLIP. Standard MACE architectures are charge-agnostic: the network sees only species and coordinates, so a foundation model trained on charge-neutral cells effectively describes V_{I}^{0}, not V_{I}^{+} — and these differ in mobility by an order of magnitude (Tyagi et al., 2025). Every ranked ΔE_a is therefore computed with a model fine-tuned *exclusively* on charged-supercell DFT data at fixed composition and charge — one model per charge state, following the validated protocol of Tyagi et al. (2025). Long-range electrostatic truncation error is monitored through supercell-size tests; should it prove limiting, a latent-Ewald long-range layer compatible with MACE (Cheng, 2025) will be added, with multi-fidelity charged-defect schemes (Wang, Mosquera-Lois and Walsh, 2026) as a further fallback.

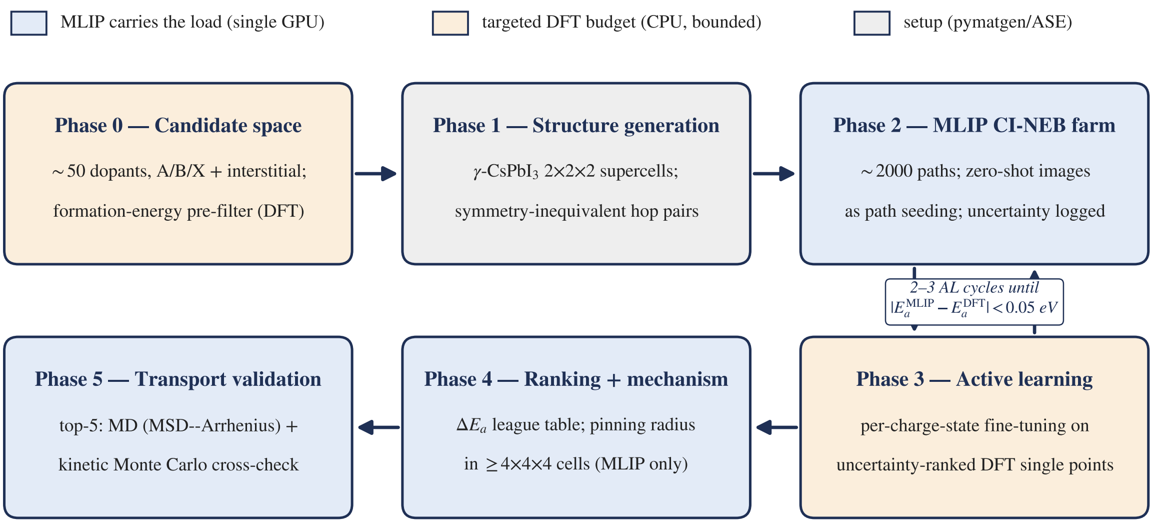

Figure 2: Screening pipeline. The MLIP (single GPU) carries >95% of force evaluations; the DFT budget (CPU) is front-loaded into validation anchors and targeted by active-learning uncertainty. Zero-shot foundation-model output is used only to seed migration paths; all ranked barriers come from per-charge-state fine-tuned models.

5 Methodology and Work Plan

5.1 Pipeline architecture

Figure 2 summarises the six-phase pipeline and the division of labour between the MLIP and the bounded DFT budget.

1. **Phase 0 — Candidate space definition.** Assemble ∼50 dopants across the four structural classes of Objective 2; pre-filter by computed dopant formation energy to exclude synthetically implausible candidates.
2. **Phase 1 — Structure generation** (pymatgen/ASE). Substitutional and interstitial injection into $2 \times 2 \times 2$ γ-$CsPbI_3$ supercells; symmetry-inequivalent configuration enumeration; generation of iodide-vacancy hop pairs binned by dopant–vacancy distance (first shell, second shell, remote bulk reference).
3. **Phase 2 — MLIP relaxation and CI-NEB farm** (ASE + MACE). Foundation-model IDPP-refined seeding of seven images; FIRE optimisation to $f_{\max} < 0.03$ eV Å^{-1}; per-path uncertainty logging.
4. **Phase 3 — Active-learning closure** (2–3 cycles). Uncertainty-ranked *charged-supercell* DFT single points (VASP/Quantum ESPRESSO); training sets mixed across endpoints, intermediate images, and perturbed configurations to prevent saddle-point overfitting; per-charge-state fine-tuning; re-run. The ranked pre-screen of Phase 4 uses fine-tuned models only.
5. **Phase 4 — Ranking and mechanistic attribution.** ΔE_a league table from the fine-tuned models; pinning-radius curves $\Delta E_a(d_{\text{dopant–vacancy}})$ extracted in $\geq 4 \times 4 \times 4$ supercells; saddle-point fingerprints (bond lengths, hydrogen-bond counts, Bader charges) mapped onto the three mechanism classes; an HSE-level density-of-states check flags dopant-induced gap states, so the final table ranks candidates that pin the lattice *without* poisoning the optoelectronics.

6. **Phase 5 — From barriers to transport.** For the top five candidates, MLIP molecular dynamics ($\geq$10 ns, mean-square-displacement Arrhenius analysis) provides a TST-free cross-check of ΔE_a, benchmarked against the $\sim$80 ns MD standard set by Arber et al. (2025); kinetic Monte Carlo over the computed rate catalogue then yields macroscopic diffusion coefficients from two independent routes.

5.2 Twelve-month timeline

Months	Activity	Milestone
1–2	Minimal viable pipeline: undoped γ-$CsPbI_3$, single V_{I} CI-NEB, zero-shot path seeding, DFT convergence and charged-defect correction budget	Pipeline runs end-to-end; E_a within literature range
3–4	Objective 1 anchors: charge-state ordering vs Tyagi et al.; GA^+ ΔE_a; strain–E_a curve under imposed lattice strain	Four anchors reproduced
5–7	Phases 1–2 at scale: full candidate enumeration; MLIP-NEB farm ($\sim$2000 paths) with first-cycle fine-tuned models	Fine-tuned ΔE_a pre-screen
8–9	Phase 3: active-learning closure; per-charge-state model validation	$\lvert E_a^{\mathrm{MLIP}} - E_a^{\mathrm{DFT}} \rvert < 0.05$ eV
10–11	Phase 4: final ranking, pinning radii in large supercells, mechanism fingerprints; Phase 5 for top five	Mechanism-annotated league table
12	Write-up; code release and documentation	Dissertation + open-source repository

Table 1: Work plan. The DFT budget is concentrated in months 1–4 and 8–9; the screening burden (months 5–7) is carried almost entirely by the MLIP.

5.3 Validation strategy

Every layer carries an internal check: supercell-size extrapolation (Layer 4), SOC-sensitivity tests on representative barriers (Layer 4), HSE spot checks on top-ranked candidates (Layer 4), MLIP-vs-DFT barrier validation on a held-out path set (Layer 5), TST-free MD cross-validation of the top five candidates (Phase 5), and, at the physics level, the four anchors of Objective 1 — three experimental, one computational. No new-dopant prediction is reported unless all anchors are reproduced.

6 Risk Assessment

Table 2 ranks the principal risks by severity. Both High-severity risks carry *structural* mitigations already embedded in the pipeline design — restriction of zero-shot output to path seeding, and per-charge-state fine-tuning — rather than contingencies bolted on after the fact.

Risk	Severity	Mitigation
MLIP extrapolation failure at saddle points	High	Zero-shot output restricted to path seeding; active learning with uncertainty sampling; mandatory DFT spot checks on all top-ranked candidates
Charge-state misrepresentation in the MLIP	High	Per-charge-state fine-tuning on charged-supercell data (Tyagi et al., 2025); FNV corrections throughout; benchmark against published V_I values
Long-range electrostatics beyond MLIP cutoff	Medium	Supercell-size monitoring; latent-Ewald augmentation (Cheng, 2025) or multi-fidelity schemes (Wang et al., 2026) as fallback
Bulk-only scope vs GB-dominated transport in films	Medium	Bulk ΔE_a sets the intrinsic floor for hysteresis and drift; grain-boundary supercells are a natural pipeline extension (Pols et al., 2024)
Organic-cation orientational sampling	Medium	Method development on $CsPbI_3$; orientation-averaged barriers for hybrid compositions
Finite-size effects	Medium	Supercell convergence tests with extrapolation; two-tier small/large-cell design
Single-hop picture misses collective effects	Medium	Phase 5 kMC/MD cross-validation
Candidate dopants synthetically inaccessible	Medium	Formation-energy pre-filter in Phase 0
Migration-optimal dopant degrades optoelectronic performance	Medium	HSE density-of-states benignity gate in Phase 4 (band edges are Pb–I-derived, so A-site and interstitial dopants are low-risk a priori); pinning radius minimises the required concentration
HPC allocation shortfall	Low	MLIP carries >95% of force evaluations; DFT budget is bounded and front-loaded

Table 2: Principal risks and mitigations.

7 Expected Outcomes and Impact

1. **A mechanism-annotated dopant ranking** — the first systematic ΔE_a league table at the ∼50-candidate scale in which every entry carries a saddle-point-resolved mechanistic label, directly answering the field's acknowledged inability to specify which defect species each strategy immobilises.

2. **The pinning radius as a new design metric** — the decay length of ΔE_a with dopant–vacancy separation determines the minimum dopant concentration for lattice-wide protection, $c_{\min} \sim (a/2r_{\text{pin}})^3$, a quantity inaccessible to experiment but decisive for formulation, and the quantitative bridge between simulation-cell dopant fractions and experimental trace-doping levels. Because every adverse electronic side effect of doping — bandgap shift, band tailing, secondary phases — grows monotonically with concentration, r_{pin} is simultaneously the performance-preservation metric: the best dopant protects the lattice at the lowest electronic cost.

3. **Transferable open infrastructure** — the released pipeline generalises immediately to bromide and mixed-halide systems, tin-based perovskites, grain-boundary supercells, and, beyond photovoltaics, to any vacancy-mediated transport problem (solid electrolytes, memristors).

4. **Experimental hand-off** — the top three to five candidates constitute concrete, falsifiable synthesis targets for collaborating device laboratories, with temperature-dependent conductivity as the direct experimental test of the predicted ΔE_a; every handed-off candidate is certified both kinetically (HSE-verified barrier) and electronically (no dopant-induced gap states).

8 Feasibility

The candidate brings (i) hands-on device-fabrication context from a p–i–n perovskite fabrication and characterisation programme (HTL/SAM, perovskite, ETL and electrode deposition; XRD, SEM, J–V/EQE and maximum-power-point stability measurement), grounding the computational choices in device reality — notably the buried-interface strain relevance of the SAM layer; (ii) prior experience with the exact machine-learning methodology required, including graph-neural-network training with active-learning cycles (validation MAE 0.067 ppm in an NMR chemical-shift project) and Bayesian-optimisation experiment design; and (iii) working familiarity with the ASE/pymatgen/MACE software stack. The project is further de-risked by design: the months 1–4 milestones deliberately reproduce published results (Eames et al., 2015; Tyagi et al., 2025), so the novel screening campaign launches from a validated baseline rather than an untested pipeline. Compute requirements are modest: the MLIP carries the screening load on a single GPU, while the bounded DFT budget ($\sim 10^2$ single-point batches plus ~ 20 full NEB benchmarks) fits within a standard university HPC allocation.

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